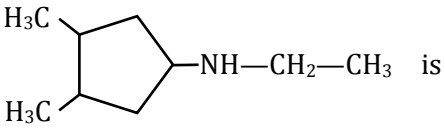
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Classification, Introduction, Preparation and Physical properties of amines

1. The IUPAC name of the amine is:



(A) N – Ethyl-2, 3-dimethyl cyclopentanamine
(B) N-ethyl-3, 4-dimethyl cyclopentanamine
(C) N-3, 4-dimethyl cyclopentanamine ethanamine
(D) N-Ethyl-3, 3-dimethyl cyclopentanamine.

CCN096

2. On reduction of Schiff's base we get :-

(A) Primary amine (B) Secondary amine
(C) Aniline (D) Anilide

CCN097

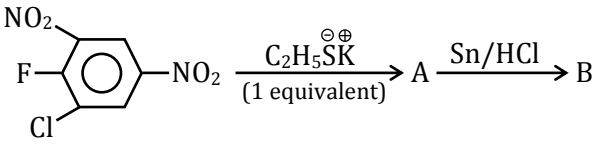
3. Among the following isomeric amines, which one is expected to have the lowest boiling point:-

(A) $CH_3(CH_2)_3NH_2$ (B) $(CH_3CH_2)_2NH$
(C) $(CH_3)_2CHNHCH_3$ (D) $CH_3CH_2N(CH_3)_2$

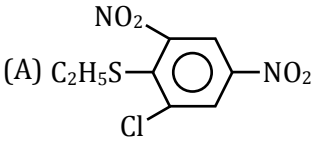
Kalinters Academy JEE

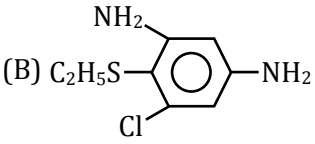
CCN098

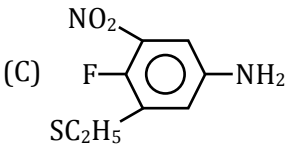
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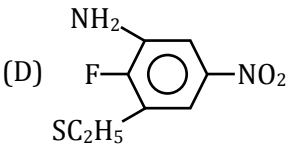


In the above reaction sequence B is :

(A) 

(B) 

(C) 

(D) 

CCN099

Chemical Reaction of Organic Compounds Containing Nitrogen

5. Benzylamine and aniline can be distinguished by

(A) Isocynide test (B) Musturd oil test
(C) Dye-azo test (D) Hinsberg test

CCN100

For More Join: @IITJEE_Advanced

6. Consider the following reaction sequence

$A \xrightarrow{H_2/Ni} B \xrightarrow[HCl]{NaNO_2} C_2H_5OH$

The starting compound (A) is :-

(A) CH_3CN (B) CH_3NC (C) C_2H_5CN (D) CH_3NO_2

CCN101

7. Consider the following sequence of reactions :-

$X(C_3H_9N) \xrightarrow{HNO_2} C_3H_8O \xrightarrow[H_2SO_4]{K_2Cr_2O_7} C_3H_6O_2$

The compound (X) is -

(A) $CH_3CH_2NHCH_3$ (B) $CH_3CH_2CH_2NH_2$ (C) $(CH_3)_2CHNH_2$ (D) $(CH_3)_3N$

CCN102

8. Choose the wrong statement :

(A) 1°Amine gives mustard oil reaction

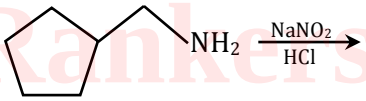
(B) 1° Amine forms salt with H_2PtCl_6

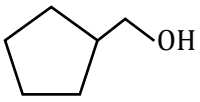
(C) 1° Amine gives hydrogen with sodium

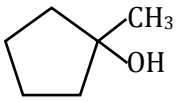
(D) 1° Amine gives alcohol by hydrolysis

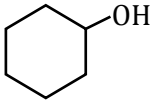
CCN103

9. The major product formed in the reaction :



(A) 

(B) 

(C) 

(D) All of these

CCN104

10. In which of the following reaction cyanide will be obtained as a major product:

(A) $Ph-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow[(ii)H_3O^{\oplus}]{(i)LiAlH_4}$

(B) $Ph-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[Br_2]{NaOH}$

(C) $Ph-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[\Delta]{P_4O_{10}}$

(D) $Ph-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{SOCl_2} \xrightarrow{NH_3}$

CCN105

11. Which of the following orders regarding the basic strength of substituted aniline is correct ?

(A) p–methylaniline > p–chloroaniline > p–aminoacetophenone

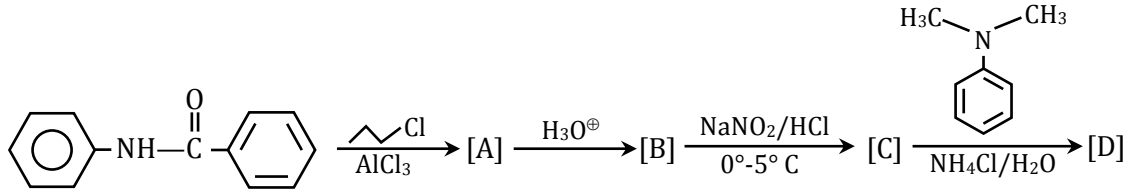
(B) p–methylaniline > p–aminoacetophenone > p–chloroaniline

(C) p–aminoacetophenone > p–methylaniline > p–chloroaniline

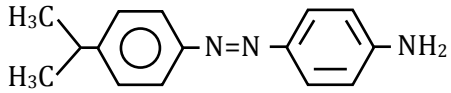
(D) p–aminoacetophenone > p–chloroaniline > p–methylaniline

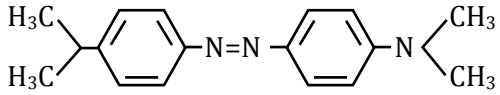
CCN106

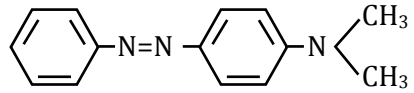
Rankers Academy JEE
For More Join: @IITJEE_Advanced

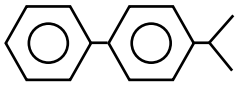
12. 

Product [D] is

(A) 

(B) 

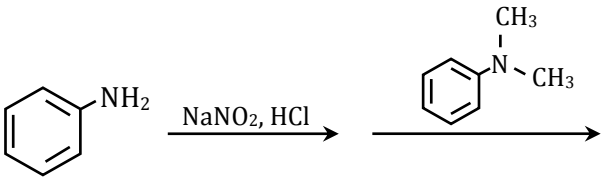
(C) 

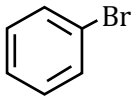
(D) 

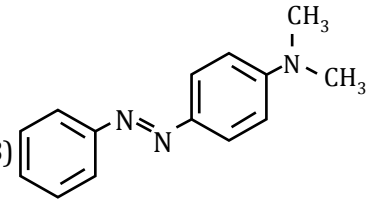
CCN107

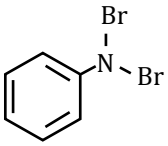
Rankers Academy JEE

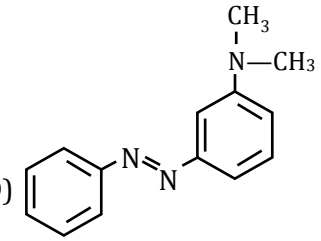
13. What could be the product for the following reaction?



(A) 

(B) 

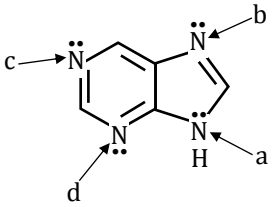
(C) 

(D) 

CCN108

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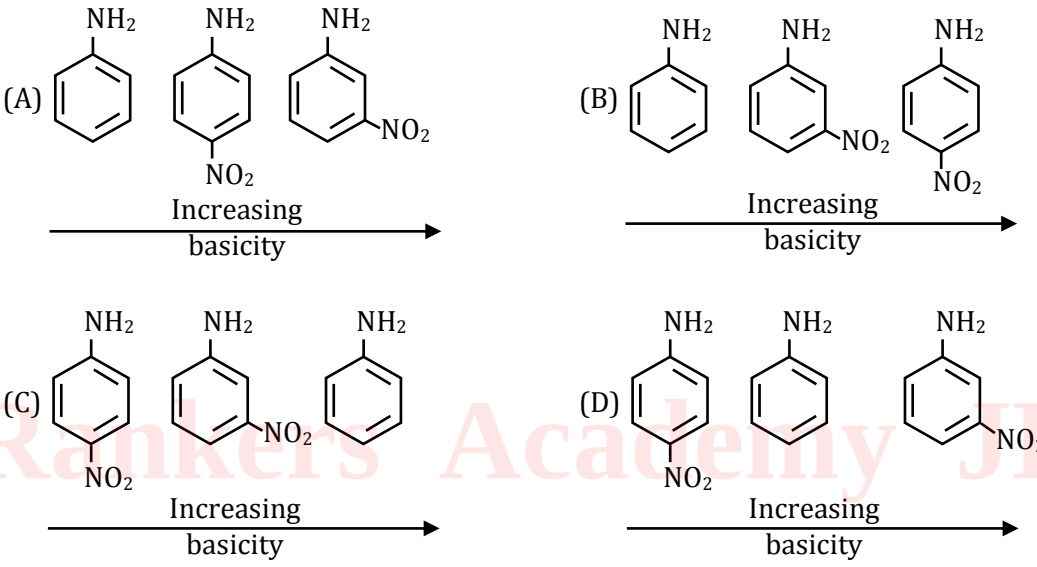
14. For the following compound, which nitrogen is most apt to be protonated?



(A) Nitrogen indicated by arrow 'a'
(B) Nitrogen indicated by arrow 'b'
(C) Nitrogen indicated by arrow 'c'
(D) Nitrogen indicated by arrow 'd'

CCN109

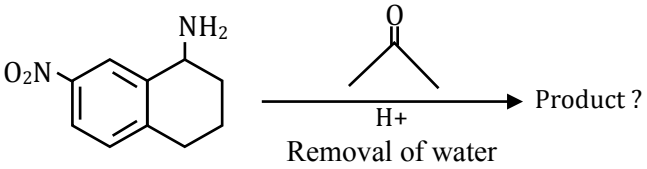
15. Choose the order the has the following compounds correctly arranged with respect to increasing basicity.

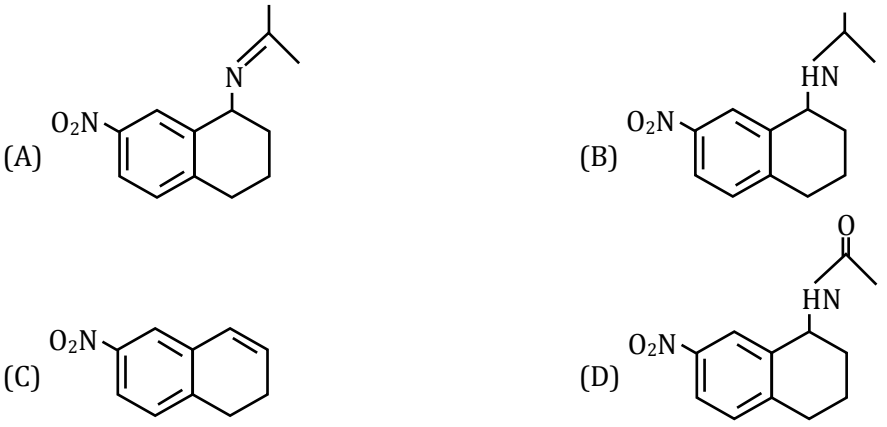


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CCN110

16. What could be the product for the following reaction?





CCN111

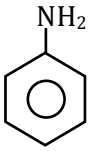
For More Join: @IITJEE_Advanced

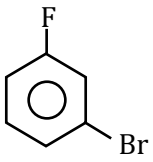
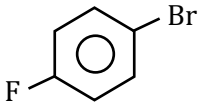
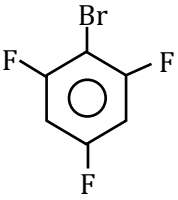
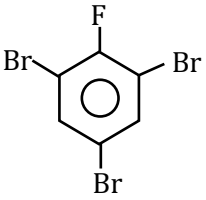
17. In the reaction, the product (C) is :

$C_6H_5NH_2 \xrightarrow[0-5°C]{NaNO_2 + HCl} (A) \xrightarrow[KCN]{CuCN} (B) \xrightarrow{H^+/H_2O} (C)$

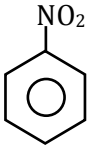
(A) $C_6H_5CH_2NH_2$ (B) C_6H_5COOH (C) C_6H_5OH (D) None of these

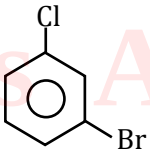
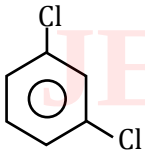
CCN161

18.  $\xrightarrow[(ii)\ HNO_2/(0-5°C)\ (iii)\ HBF_4/\Delta]{(i)\ Br_2/H_2O}$ (A). The major product (A) is

(A)  (B)  (C)  (D) 

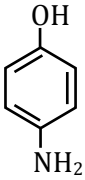
CCN162

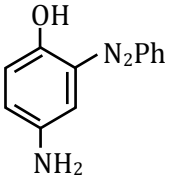
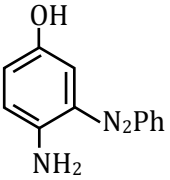
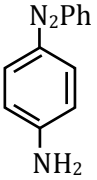
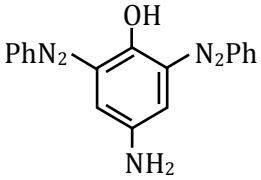
19.  $\xrightarrow{(i)\ Br_2/Fe\ (ii)\ H_2/Pt}$ (A) $\xrightarrow[(ii)\ Cu_2Cl_2/HCl]{(i)\ NaNO_2/HCl\ 0-5°C}$ (B). The product (B) is

(A)  (B)  (C)  (D) 

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CCN163

20.  + $Ph-\overset{\oplus}{N}_2Cl^{\ominus}$ $\xrightarrow{Ph = 4-6}$ X (major product) X will be -

(A)  (B)  (C)  (D) 

CCN164

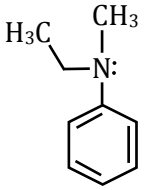
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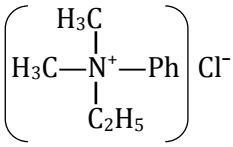
EXERCISE (O-2)

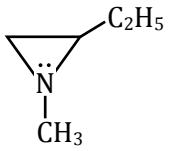
MULTIPLE CORRECT TYPE QUESTIONS

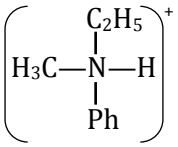
Classification, Introduction, Preparation and Physical properties of amines

1. Which of the following compund(s) is (are) resolvable?

(A) 

(B) 

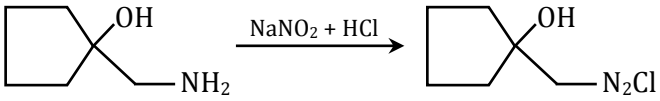
(C) 

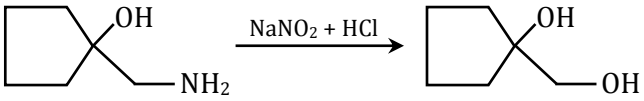
(D) 

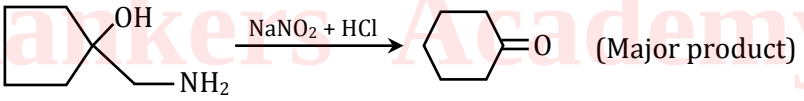
CCN112

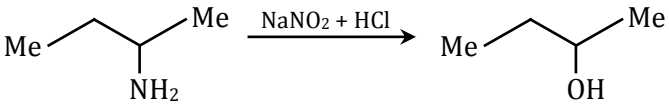
Chemical Reaction of Organic compounds containing Nitrogen

2. Which of the following reactions are correct ?

(A)  (Major product)

(B)  (Major product)

(C)  (Major product)

(D) 

CCN113

3. Which of the following statements are correct?

(A) Aryldiazonium ions are more stable than alkyldizonium ions

(B) Electron release from the ortho and para-positions of the ring stabilizes the aryldiazonium ion

(C) The increased stability of aryldiazonium is due to the great difficulty of forming $Ar^{\oplus}$

(D) Alkyldiazonium is more stable than aryldiazonium ion

CCN114

4. Mixture of 1°, 2° and 3° amines can be separated by :

(A) Hinsberg's method
(B) Hofmann's isocyanide test
(C) Fractional distillation
(D) $NaNO_2$ HCl

CCN165

5. Which of the following amine reacts with Hinsberg reagent to give base soluble product :-

(A) Neopentyl amine
(B) sec propyl amine
(C) diethyl amine
(D) ethyl methyl amine

CCN166

For More Join: @IITJEE_Advanced

6. $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_2-CH_3 \xrightleftharpoons{H^{18}OH/H^{\oplus}}$ Ethanoic acid + Ethanol

Isotopic oxygen of water will be present with

(A) Ethanoic acid

(B) Ethanol

(C) After some time it will also be present in some molecules of ester

(D) None of these

CCN167

7. The presence of alphatic primary amine can be confirmed by its reaction with :

(A) HNO_2 followed by CAN solution (B) $CHCl_3$ + NaOH

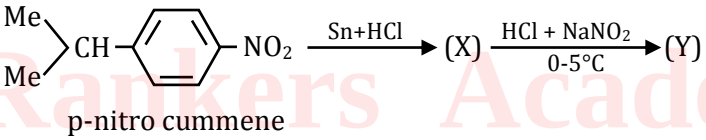
(C) CS_2 & $HgCl_2$ (D) H_2SO_4

CCN168

COMPREHENSION TYPE QUESTIONS

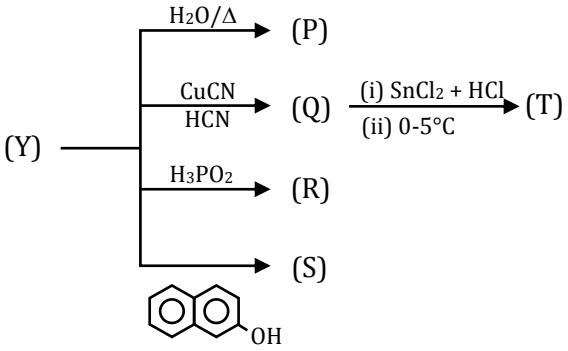
Chemical Reaction of Organic compounds containing Nitrogen

Paragraph for Question No. 8 to 9



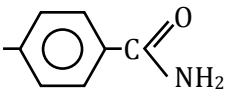
p-nitro cummene

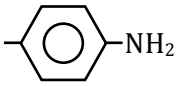
Rankers Academy JEE

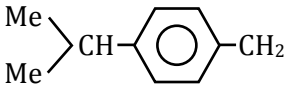


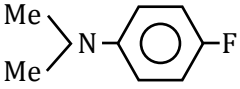
For the given reaction sequence, answer the following

8. $(Y) \xrightarrow[\Delta]{HBF_4} \xrightarrow[\Delta]{KMnO_4/H^{\ominus}} \xrightarrow[H_2SO_4]{N_3H}$ Final product; final product is

(A) 

(B) 

(C) 

(D) 

CCN115

For More Join: @IITJEE_Advanced

9. $(R) \xrightarrow[(ii)\ NaOH + I_2]{(i)\ CH_3COCl/AlCl_3} (A) + \underset{\text{Yellow ppt.}}{(B)\downarrow}$

$$(A) \xrightarrow[(iii)\ (R)/AlCl_3]{(i)\ H_2O/H^{\oplus}\ (ii)\ SOCl_2} (L)$$

The correct statement about the product (L) is

(A) Product (L) on reaction with Tollen's reagent gives silver mirror

(B) Product (L) on reaction with $H_2N\text{-}NH_2/^-OH, \Delta$ gives hydrocarbon as product

(C) Product (L) on reaction with HCN gives a compound which contains a chiral centre

(D) All of these

CCN116

Paragraph for Question No. 10 to 12

The isomeric compounds 'A' and 'B' with molecular formula C_7H_9N gave the following reactions. When B is acetylated and then brominated, it gave one monobromo derivative. 'A' when treated with $NaNO_2$ and HCl gave the compound 'C'. 'C' was heated with acetic acid in the presence of conc. H_2SO_4, a pleasant smelling liquid (D) was obtained. 'B' was treated with $NaNO_2$ and HCl in cold condition and treated with β-Naphthol in NaOH to give an orange red dye.

10. 'B' is

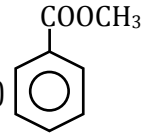
(A) benzylamine (B) p-toluidine

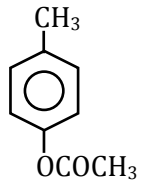
(C) o-toluidine (D) m-toluidine

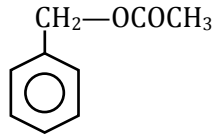
Pankers Academy JEE

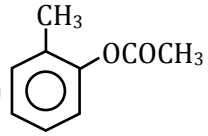
CCN118

11. 'D' is

(A) 

(B) 

(C) 

(D) 

CCN119

12. 'A' is

(A) p-toluidine (B) N-methylaniline

(C) p-toluidine (D) benzylamine

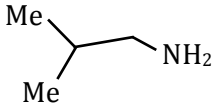
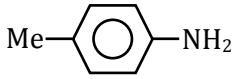
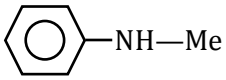
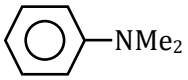
CCN120

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

Chemical Reaction of Organic compounds containing Nitrogen

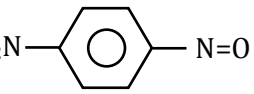
13. Match the following

List-I		List-II	
(A)		(p)	Liebermann's nitroso reaction
(B)		(q)	Evolution of N_2 with HNO_2
(C)		(r)	Dye test after diazotization
(D)		(s)	Green colour with HNO_2
		(t)	Carbylamine test

(A) (A – q, r) ; (B – r, t) ; (C – p) ; (D – s)
(B) (A – q, t) ; (B – r, t) ; (C – p) ; (D – s)
(C) (A – q, t) ; (B – r, s) ; (C – p) ; (D – s)
(D) (A – q, t) ; (B – r, t) ; (C – q) ; (D – s)

CCN121

14.

Column-I		Column-II	
(A)	$R'NH_2 \xrightarrow{NaNO_2+HCl}$	(P)	RR′N–N=O
(B)	$PhNH_2 \xrightarrow[0°-5°C]{NaNO_2+HCl}$	(Q)	$Ph-\overset{\oplus}{N}\equiv NCl^{\ominus}$
(C)	$PhN(CH_3)_2 \xrightarrow{NaNO_2+HCl}$	(R)	
(D)	$RR'NH \xrightarrow{NaNO_2+HCl}$	(S)	R′OH + N_2

(A) A → S, B → Q, C → R, D → P
(B) A → Q, B → S, C → R, D → P
(C) A → P, B → Q, C → R, D → S
(D) A → S, B → R, C → P, D → Q

CCN122

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For More Join: @IITJEE_Advanced

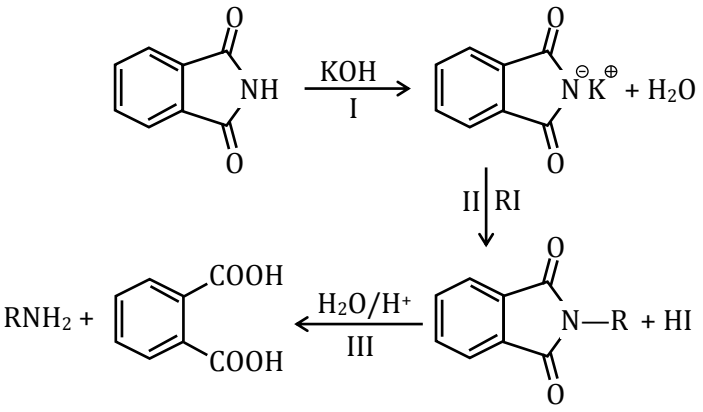
EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Classification, Introduction, Preparation and Physical properties of amines

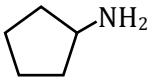
1. Gabriel's phthalimide reaction can be out lined as follows

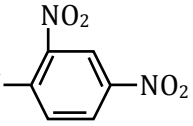


How many of the following amines can be prepared by this method

(i) $H_3C—\underset{\displaystyle NH_2}{\underset{|}{CH}}—CH_3$ (ii) $CH_3NH—CH_2CH_3$

(iii) $C_6H_5NH_2$ (iv) $C_6H_5CH_2NH_2$

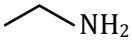
(v) 

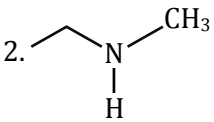
(vi) 

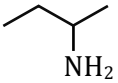
CCN123

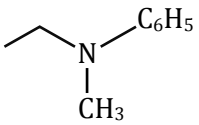
Chemical Reaction of Organic compounds containing Nitrogen

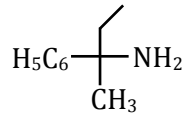
2. Of the following amines how many can be separated by Hoffmann's mustard oil reaction?

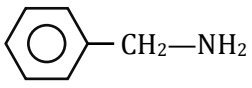
1. 

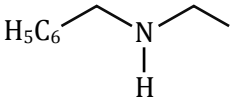
2. 

3. 

4. 

5. 

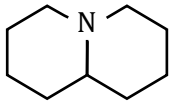
6. 

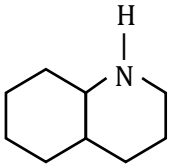
7. 

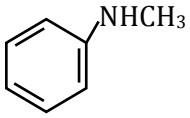
CCN124

For More Join: @IITJEE_Advanced

3. How many of the following gives visible change when treated with nitrous acid ($NaNO_2$ + HCl)?

(i) $CH_3CH_2CH_2NH_2$ (ii) $CH_2CH_2NHCH_3$ (iii) 

(iv)  (v) $CH_3CH_2CONH_2$ (vi) $CH_3CH_2CH(CH_3)NO_2$

(vii) $CH_3CH_2N(CH_3)_2$ (viii)  (ix) $CH_3COOC_2H_5$

CCN125

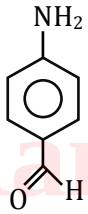
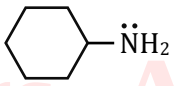
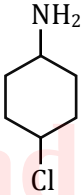
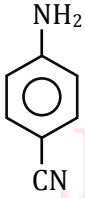
4. When α-amine propanoic acid is heated, how many isomeric piperazines (including stereoisomers are obtained).

CCN126

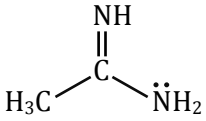
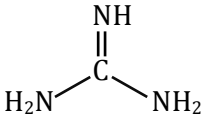
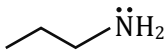
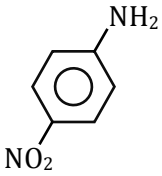
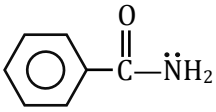
5. When β-aminobutanoic acid is heated how many diastereomers are obtained

CCN127

6. Identify how many compounds are more basic than aniline.

1. 
2. 
3. 
4. 

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5. 
6. 
7. 
8. 
9. 

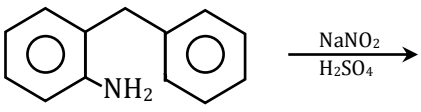
CCN128

7. Number of amines (only structural isomers) of molecular $C_5H_{13}N$ that gives yellow oily liquid on reaction with nitrous acid is ?

CCN129

8. Diazotization of n–Butylamine with $NaNO_2$/HCl gives How many number of isomeric Butene?

CCN130

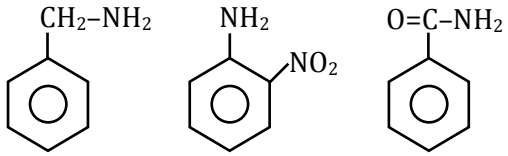
9. 

Number of ring in product (P) is x and number of π–bond is y then x+y is .

CCN131

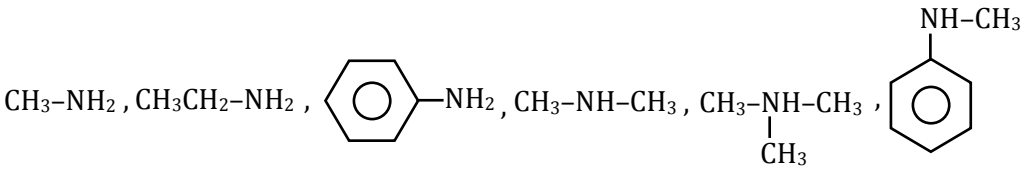
For More Join: @IITJEE_Advanced

10. How many number of compounds is/are less basic than aniline in the given compounds ?



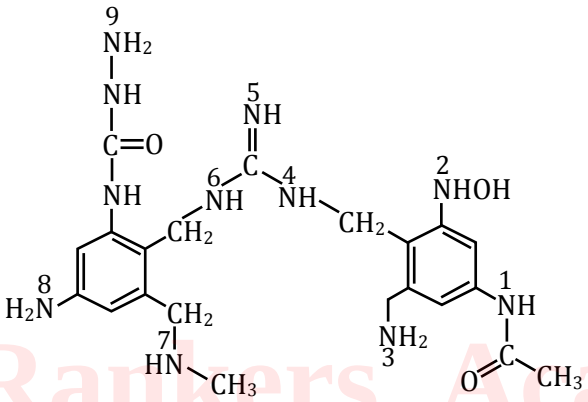
CCN132

11. How many of the following amines reacts with benzene sulfonyl chloride and in presence of KOH water soluble potassium salt are formed.



CCN133

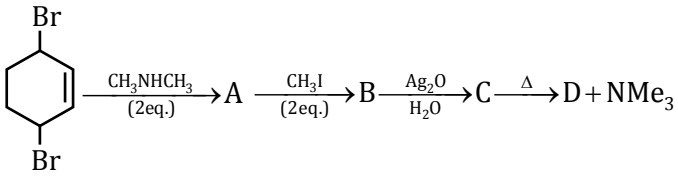
12.



The most basic N in the given molecule is ?

CCN134

13.



Number of sp^2 hybridised carbon present in D is?

CCN169

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Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Classification, Introduction, Preparation and Physical properties of amines

14. The Number of isomers with pyridine nucleus possible for compounds having Molecular Formula C_6H_7N is (Z) then $\left(\frac{Z}{2}\right)$ is

CCN170

For More Join: @IITJEE_Advanced

15. Number of aromatic (six membered carbocyclic) isomer possible for Molecular Formula C_7H_9N is y then $\left(\frac{y}{2}\right)$ is

CCN171

16. Total number of the following statements which are correct for trimethylamine is x then $\frac{x}{4}$

(1) It has a trigonal pyramidal shape.
(2) The C–N–C bond angle is 108.7°.
(3) The N–atom is sp^3 hybridized.
(4) The unshared pair of electrons on nitrogen occupies an sp^2 orbital.

CCN172

Chemical Reaction of Organic compounds containing Nitrogen

17. Number of compounds of formula $C_4H_{11}N$ give positive carbylamine reaction or test is a then $\left(\frac{a^2}{2}\right)$ is (including stereoisomer) ?

CCN173

18. The degree of unsaturation of final product formed in the reaction is Z then $\left(\frac{Z}{2}\right)$ is

$$Ph\text{–}NH_2 \xrightarrow[0\text{–}5^\circ C]{NaNO_2+HCl} \xrightarrow{Ph\text{–}OH} \text{Organic product}$$

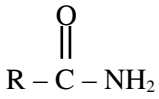
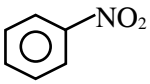
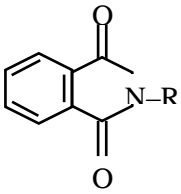
CCN174

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EXERCISE - JEE (Main) PYQ

1. Match the Compounds **(List-I)** with the appropriate Catalyst/Reagents **(List-II)** for their reduction into corresponding amines. **[JEE (Main) 2025]**

	List-I (Compounds)		**List-II (Catalyst/Reagents)**
(A)		(I)	NaOH (aqueous)
(B)		(II)	H_2/Ni
(C)	R–C≡N	(III)	$LiAlH_4$, H_2O
(D)		(IV)	Sn, HCl

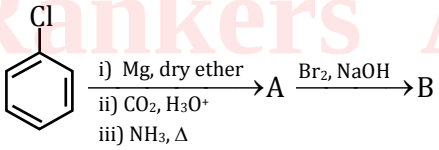
Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I) (2) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

(3) (A)-(II), (B)-(I), (C)-(III), (D)-(IV) (4) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

CCN206

2. Consider the following sequence of reactions : **[JEE (Main) 2025]**



Chlorobenzene

11.25 mg of chlorobenzene will produce ____ × 10^{-1} mg of product B.

(Consider the reactions result in complete conversion.)

[Given molar mass of C, H, O, N and Cl as 12, 1, 16, 14 and 35.5 g mol^{-1} respectively]

CCN207

3. Identify correct statements : **[JEE (Main) 2025]**

(A) Primary amines do not give diazonium salts when treated with $NaNO_2$ in acidic condition.

(B) Aliphatic and aromatic primary amines on heating with $CHCl_3$ and ethanolic KOH form carbylamines.

(C) Secondary and tertiary amines also give carbylamine test.

(D) Benzenesulfonyl chloride is known as Hinsberg's reagent.

(E) Tertiary amines reacts with benzenesulfonyl chloride very easily.

Choose the correct answer from the options given below :

(1) (B) and (D) only (2) (A) and (B) only

(3) (D) and (E) only (4) (B) and (C) only

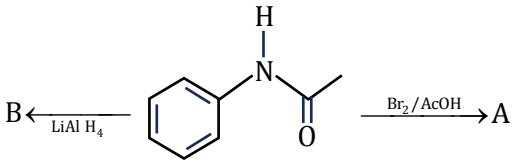
CCN208

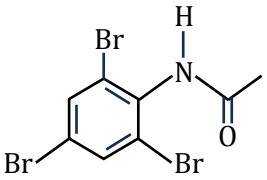
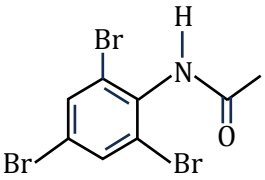
4. 9.3 g of pure aniline is treated with bromine water at room temperature to give a white precipitate of the product 'P'. The mass of product 'P' obtained is 26.4 g. The percentage yield is ..... %. **[JEE (Main) 2024]**

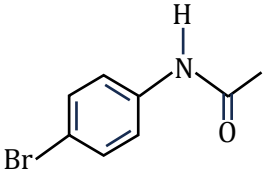
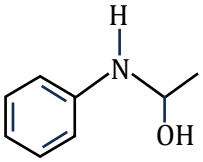
CCN175

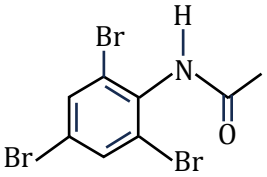
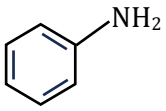
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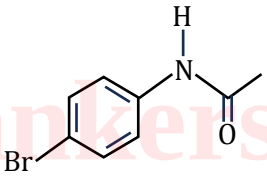
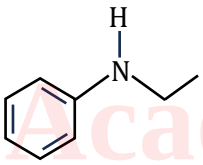
5. The major products A and B from the following reactions are: **[JEE (Main) 2023]**



(1) A =  , B = 

(2) A =  , B = 

(3) A =  , B = 

(4) A =  , B = 

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CCN137

6. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Experimental reaction of CH_3Cl with aniline and anhydrous $AlCl_3$ does **not** give o and p-methylaniline.

Reason (R) : The — NH_2 group of aniline becomes deactivating because of salt formation with anhydrous $AlCl_3$ and hence yields *m*-methyl aniline as the product.

In the light of the above statements, choose the most appropriate answer from the options given below : **[JEE (Main) 2022]**

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

CCN057

7. Phthalimide is made to undergo following sequence of reactions.

Phthalimide $\xrightarrow[\text{(ii) Benzylchloride}]{\text{(i) KOH}}$ 'P'

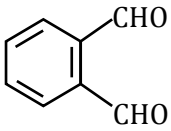
Total number of π bonds present in product 'P' is/are **[JEE (Main) 2024]**

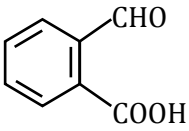
CCN176

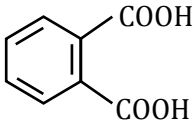
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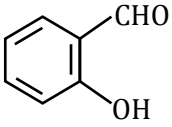
8. An organic compound 'A' on reaction with NH_3 followed by heating gives compound B. Which on further strong heating gives compound C ($C_8H_5NO_2$). Compound C on sequential reaction with ethanolic KOH, alkyl chloride and hydrolysis with alkali gives a primary amine. The compound A is :

[JEE (Main) 2022]

(1) 

(2) 

(3) 

(4) 

CCN055

9. X g of ethylamine is subjected to reaction with $NaNO_2$/HCl followed by water; evolved dinitrogen gas which occupied 2.24 L volume at STP.

X is _____ $\times 10^{-1}$ g.

[JEE (Main) 2024]

CCN177

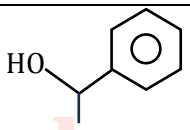
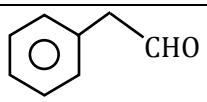
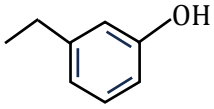
10. X gram of ethanamine was subjected to reaction with $NaNO_2$/HCl followed by hydrolysis to liberate N_2 and HCl. The HCl generated was completely neutralised by 0.2 moles of NaOH. X is___ g.

[JEE (Main) 2024]

CCN178

11. Match List-I with List-II :

[JEE (Main) 2023]

	List-I (Reagents used)		**List-II (Compound with functional group detected)**
A.	Alkaline solution of copper sulphate and sodium citrate	I.	
B.	Neutral $FeCl_3$ solution	II.	
C.	Alkaline chloroform solution	III	
D.	Potassium iodide and sodium hypochlorite	IV	

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-III, D-I

(2) A-IV, B-I, C-II, D-III

(3) A-III, B-IV, C-I, D-II

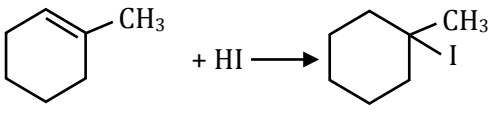
(4) A-III, B-IV, C-II, D-I

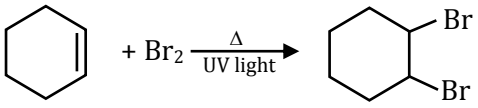
CCN140

12. Which of the following reaction is correct ?

[JEE (Main) 2024]

(1) $CH_3CH_2CH_2NH_2 \xrightarrow[H_2O]{HNO_2, 0^oC} CH_3CH_2OH + N_2 + HCl$

(2) 

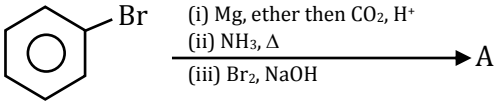
(3) 

(4) $C_2H_5CONH_2 + Br_2 + NaOH \rightarrow C_2H_5CH_2NH_2 + Na_2CO_3 + NaBr + H_2O$

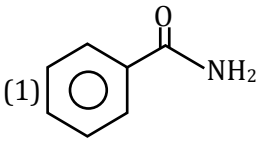
CCN179

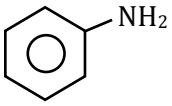
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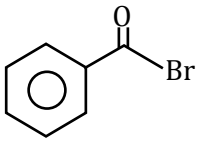
13. The final product A, formed in the following multistep reaction sequence is:

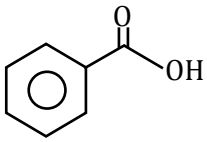


[JEE (Main) 2024]

(1) 

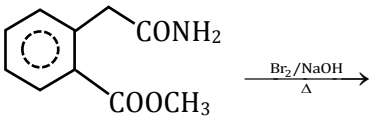
(2) 

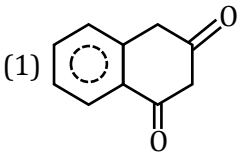
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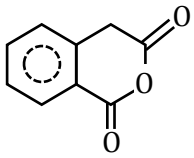
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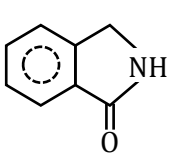
CCN180

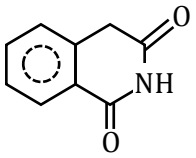
14. The major product formed in the following reaction is [JEE (Main) 2023]



(1) 

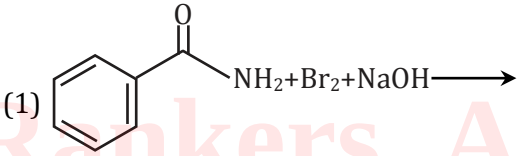
(2) 

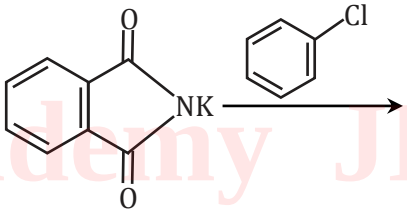
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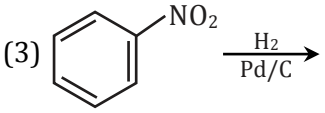
(4) 

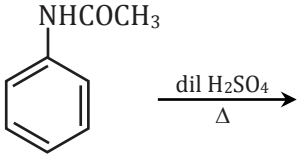
CCN138

15. How many of the transformation given below would result in aromatic amines? [JEE (Main) 2023]

(1) 

(2) 

(3) 

(4) 

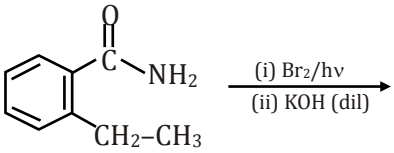
CCN142

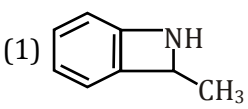
16. Reaction of propanamide with Br_2 / KOH (aq) produces : [JEE (Main) 2023]

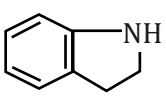
(1) Ethylnitrile (2) Propylamine (3) Propanenitrile (4) Ethylamine

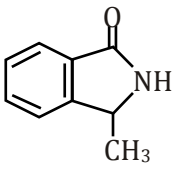
CCN143

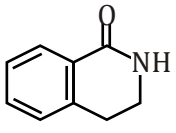
17. The major product of the following reaction is: [JEE (Main) 2019]



(1) 

(2) 

(3) 

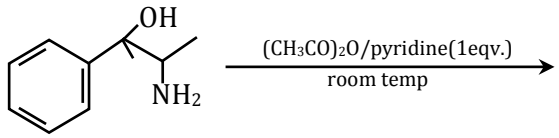
(4) 

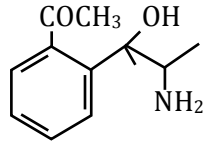
CCN135

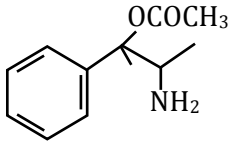
Rankers Academy JEE

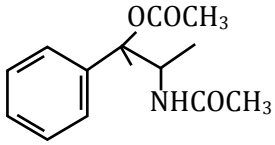
For More Join: @IITJEE_Advanced

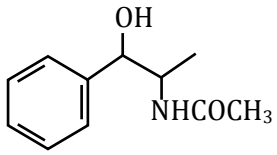
18. The major product obtained in the following reaction is:– **[JEE (Main) 2019]**



(1) 

(2) 

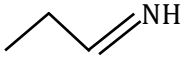
(3) 

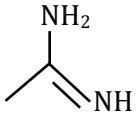
(4) 

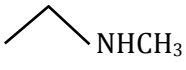
CCN041

19. The increasing order of basicity of the following compounds is : **[JEE (Main) 2018]**

(a) 

(b) 

(c) 

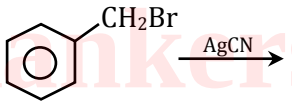
(d) 

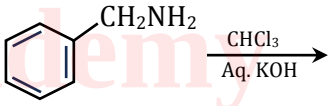
(1) (b) < (a) < (c) < (d)
(2) (b) < (a) < (d) < (c)
(3) (d) < (b) < (a) < (c)
(4) (a) < (b) < (c) < (d)

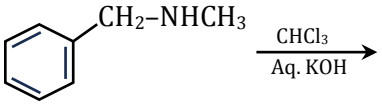
CCN039

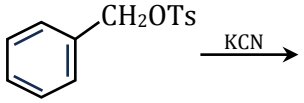
20. Benzyl isocyanide can be obtained by: **[JEE (Main) 2023]**

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(A) 

(B) 

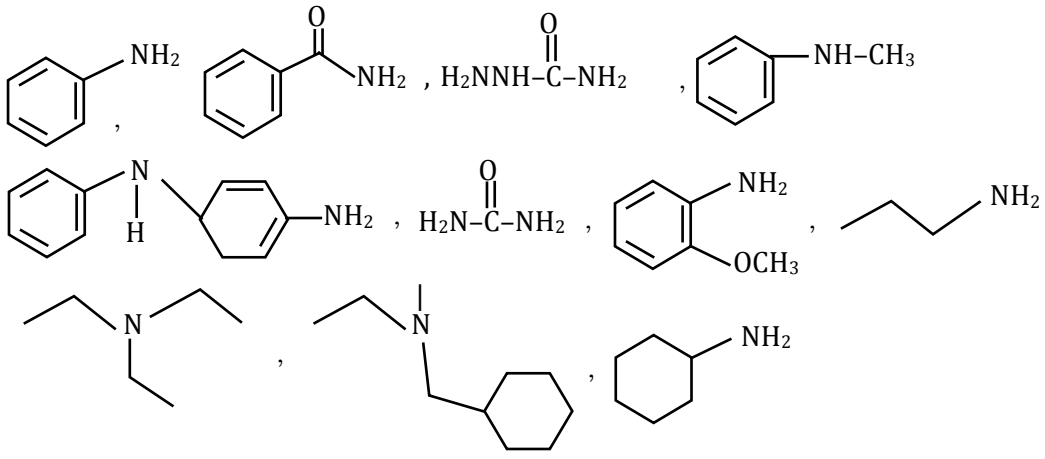
(C) 

(D) 

Choose the correct answer from the options given below :
(1) A and D (2) Only B (3) A and B (4) B and C

CCN144

21. Number of amine compounds from the following giving solids which are soluble in NaOH upon reaction with Hinsberg's reagent is ____. **[JEE (Main) 2024]**



CCN181

For More Join: @IITJEE_Advanced

22. Given below are two statements : **[JEE (Main) 2024]**

Statement (I) : All the following compounds react with p-toluenesulfonyl chloride.

$C_6H_5NH_2$ $(C_6H_5)_2NH$ $(C_6H_5)_3N$

Statement (II) : Their products in the above reaction are soluble in aqueous NaOH.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both **Statement I** and **Statement II** is false

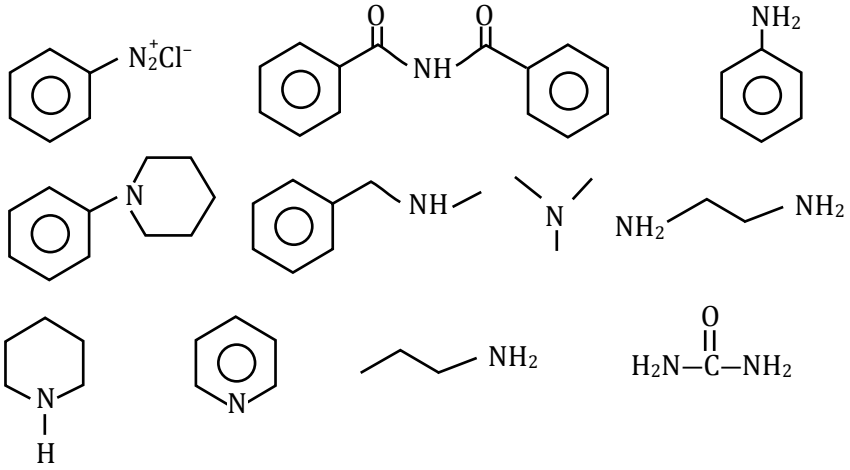
(2) **Statement I** is true but **Statement II** is false

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** is true

CCN182

23. Number of compounds which give reaction with Hinsberg's reagent is_____.



[JEE (Main) 2024]

CCN183

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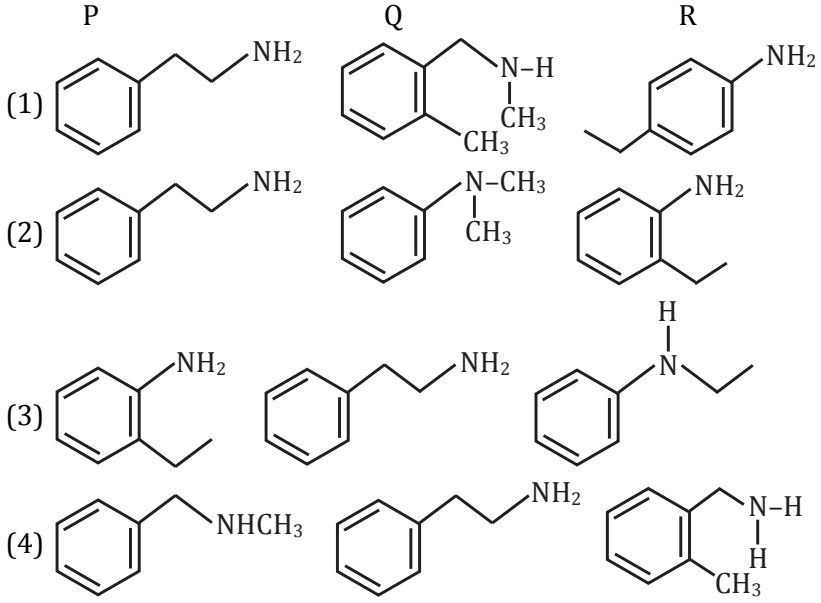
24. Isomeric amines with molecular formula $C_8H_{11}N$ give the following tests

Isomer (P) $\Rightarrow$ Can be prepared by Gabriel phthalimide synthesis

Isomer (Q) $\Rightarrow$ Reacts with Hinsberg's reagent to give solid insoluble in NaOH

Isomer (R) $\Rightarrow$ Reacts with HONO followed by β-naphthol in NaOH to give red dye.

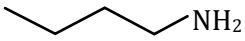
Isomers (P), (Q) and (R) respectively are **[JEE (Main) 2023]**

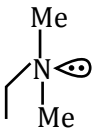


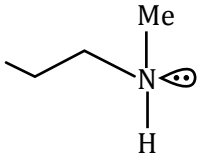
CCN139

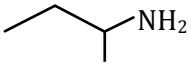
For More Join: @IITJEE_Advanced

25. An organic compound [A]($C_4H_{11}N$), shows optical activity and gives N_2 gas on treatment with HNO_2. The compound [A] reacts with $PhSO_2Cl$ producing a compound which is soluble in KOH. The structure of A is: **[JEE (Main) 2023]**

(1) 

(2) 

(3) 

(4) 

CCN141

26. Match **List I** with **List II** **[JEE (Main) 2022]**

List I	List II
A. Benzenesulphonyl chloride	I. Test for primary amines
B. Hoffmann bromamide reaction	II. Anti Saytzeff
C. Carbylamine reaction	III. Hinsberg reagent
D. Hoffmann orientation	IV. Known reaction of Isocyanates.

Choose the correct answer from the options given below:

(1) A-IV, B –III, C-II, D-I
(2) A-IV, B –II, C-I, D-III
(3) A-III, B –IV, C-I, D-II
(4) A-IV, B –III, C-I, D-II

Rankers Academy JEE

CCN058

27. Which statement is NOT correct for p-toluenesulphonyl chloride?

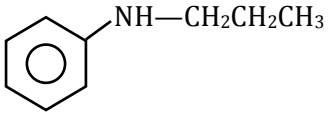
[JEE (Main) 2022]

(1) It is known as Hinsberg's reagent.
(2) It is used to distinguish primary and secondary amines.
(3) On treatment with secondary amine, it leads to a product, that is soluble in alkali.
(4) It doesn't react with tertiary amines.

CCN061

28. An amine on reaction with benzenesulphonyl chloride produces a compound insoluble in alkaline solution. This amine can be prepared by ammonolysis of ethyl chloride. The correct structure of amine is :

[JEE (Main) 2021]

(1) 

(2) $CH_3CH_2NH_2$

(3) $CH_3CH_2CH_2NHCH_3$

(4) $CH_3CH_2CH_2\overset{H}{N}—CH_2CH_3$

CCN054

For More Join: @IITJEE_Advanced

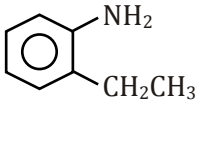
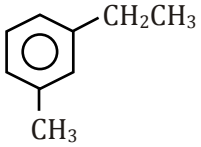
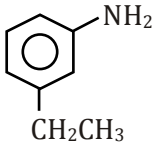
29. Three isomers A, B and C (mol. formula $C_8H_{11}N$) give the following results :

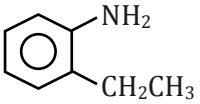
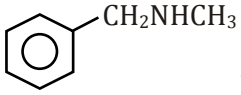
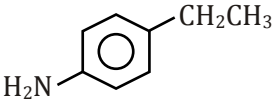
A and C $\xrightarrow{\text{Diazotization}}$ P + Q $\xrightarrow[\substack{\text{(ii) oxidation}\\(KMnO_4+H^+)}]{\text{(i) Hydrolysis}}$ R(product of A) + S(product of C)

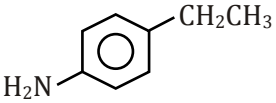
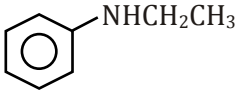
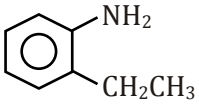
R has lower boiling point than S

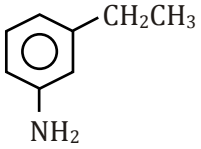
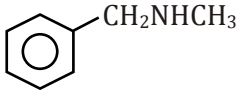
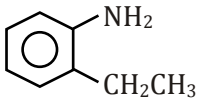
B $\xrightarrow{C_6H_5SO_2Cl}$ alkali-insoluble product

A, B and C, respectively are : **[JEE (Main) 2020]**

(1)   

(2)   

(3)   

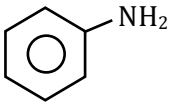
(4)   

CCN136

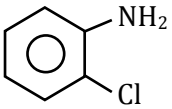
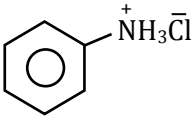
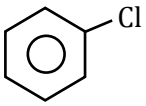
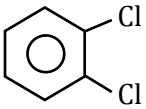
30. From 6.55 g of aniline, the maximum amount of acetanilide that can be prepared will be___$\times 10^{-1}$ g. **[JEE (Main) 2024]**

CCN184

31. The product A formed in the following reaction is:

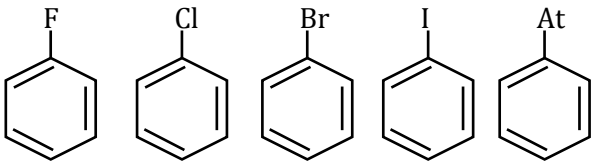
 $\xrightarrow[\text{Then } Cu_2Cl_2]{NaNO_2,\ HCl,\ 0°C}$ A

[JEE (Main) 2024]

(1)  (2)  (3)  (4) 

CCN185

32. The number of halobenzenes from the following that can be prepared by Sandmeyer's reaction is .....



I II III IV V

[JEE (Main) 2024]

CCN186

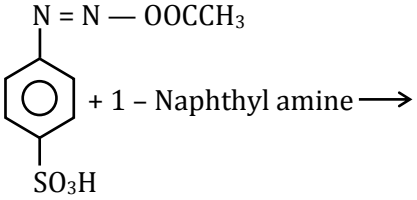
For More Join: @IITJEE_Advanced

33. 9.3 g of pure aniline upon diazotisation followed by coupling with phenol gives an orange dye. The mass of orange dye produced (assume 100% yield/ conversion) is _____g. (nearest integer)

[JEE (Main) 2024]

CCN187

34. Choose the correct colour of the product for the following reaction.



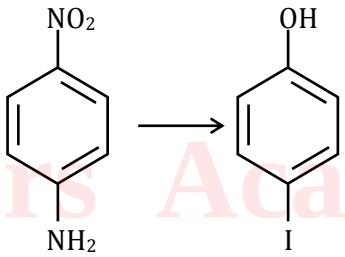
[JEE (Main) 2023]

(1) Yellow
(2) White
(3) Red
(4) Blue

CCN062

35. The correct sequential order of the reagents for the given reaction is :

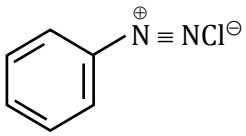
[JEE (Main) 2022]



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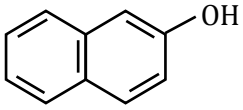
(1) HNO_2, $Fe/H^{\oplus}$, HNO_2, KI, $H_2O/H^{\oplus}$
(2) HNO_2, KI, $Fe/H^{\oplus}$, HNO_2, H_2O/warm
(3) HNO_2, KI, HNO_2, $Fe/H^{\oplus}$, $H_2O/H^{\oplus}$
(4) HNO_2, $Fe/H^{\oplus}$, KI, HNO_2, H_2O/warm

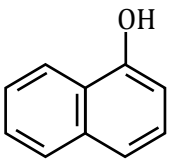
CCN056

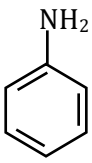
36.  + 'A' $\xrightarrow{NaOH}$ Product (Orange-red dye)

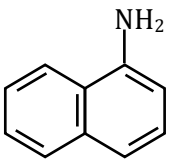
Which among the following represent reagent 'A'?

[JEE (Main) 2022]

(1) 

(2) 

(3) 

(4) 

CCN059

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Match the compounds in LIST-I with the observation in LIST-II and choose the correct option.

[JEE (Advanced) 2022]

LIST-I		LIST-II	
(I)	Aniline	(P)	Sodium fusion extract of the compound on boiling with $FeSO_4$, followed by acidification with conc. H_2SO_4, gives Prussian blue color.
(II)	o-Cresol	(Q)	Sodium fusion extract of the compound on treatment with sodium nitroprusside gives blood red color.
(III)	Cysteine	(R)	Addition of the compound to a saturated solution of $NaHCO_3$ results in effervescence.
(IV)	Caprolactam	(S)	The compound reacts with bromine water to give a white precipitate.
		(T)	Treating the compound with neutral $FeCl_3$solution produces violet color.

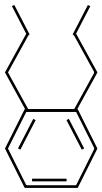
(A) I → P, Q; II → S; III → Q, R; IV → P
(B) I → P ; II → R, S; III → R; IV → Q, S
(C) I → Q, S; II → P, T; III → P; IV → S
(D) I → P, S; II → T; III → Q, R; IV → P

CCN189

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Paragraph for Question No. 2 to 3

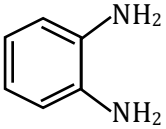
Treatment of compound **O** with $KMnO_4$ / $H^{\oplus}$ gave **P**, which on heating with ammonia gave the compound **Q** which on treatment with Br_2 / NaOH produced **R**. On strong heating, **Q** gave **S**, which on further treatmenet with ethyl 2-bromopropanoate in the presence of KOH following by acidification, gave a compound **T**.

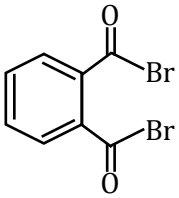


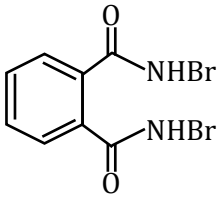
(O)

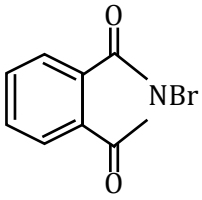
2. The compound **R** is

[JEE (Advanced) 2016]

(A) 

(B) 

(C) 

(D) 

CCN088

For More Join: @IITJEE_Advanced

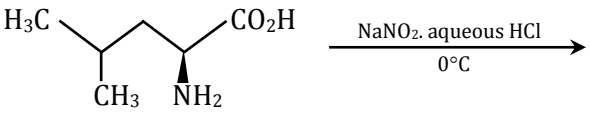
3. The compound **T** is :

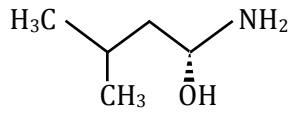
[JEE (Advanced) 2016]

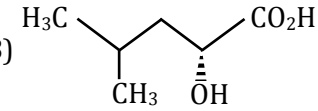
(A) Glycine (B) Alanine (C) Valine (D) Serine

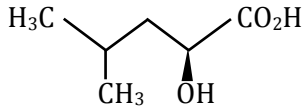
CCN089

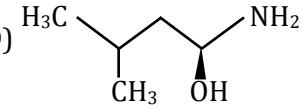
4. The major product of the reaction is: **[JEE (Advanced) 2015]**



(A) 

(B) 

(C) 

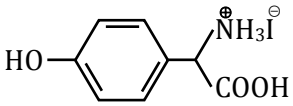
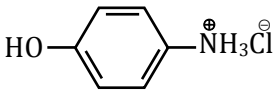
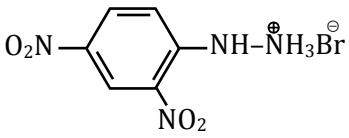
(D) 

CCN087

5. Match the compounds in **Column I** with their characteristic test(s)/reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

[IIT - 2008]

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Column-I		Column-II	
(A)	$H_2N-\overset{\oplus}{N}H_3\overset{\ominus}{Cl}$	(P)	Sodium fusion extract of the compound gives Prussian blue colour with $FeSO_4$
(B)		(Q)	Gives positive $FeCl_3$ test
(C)		(R)	Gives white precipitate with $AgNO_3$
(D)		(S)	Reacts with aldehydes to form the corresponding hydrazone derivative

(A) (A) → R,S; (B) → P, Q; (C) → P,Q,R ; (D) → P,S

(B) (A) → R; (B) → P,S; (C) → P,Q,R ; (D) → P,S

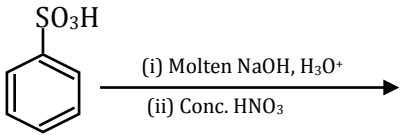
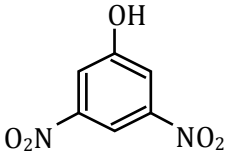
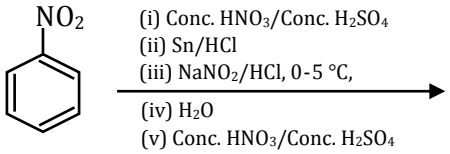
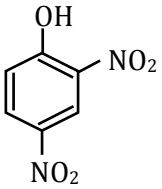
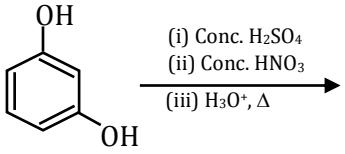
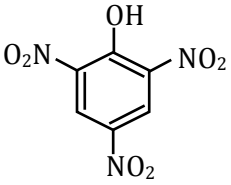
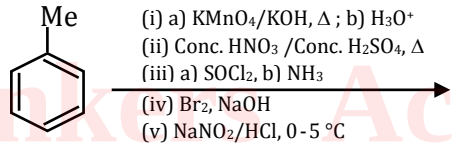
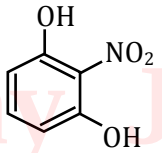
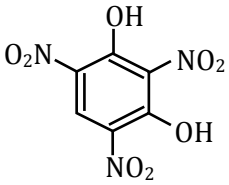
(C) (A) → S; (B) → P, Q; (C) → Q,R,S ; (D) → P,S

(D) (A) → p,q; (B) → P, R; (C) → P,Q,R ; (D) → P,S

CCN190

For More Join: @IITJEE_Advanced

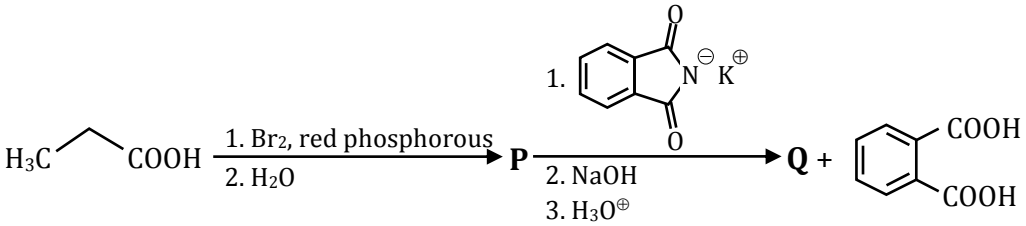
6. **List-I** contains various reaction sequences and **List-II** contains different phenolic compounds. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option. **[JEE (Advanced) 2024]**

List-I		List-II	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	
		(5)	

(A) P-2, Q-3, R-4, S-5
(B) P-2, Q-3, R-5, S-1
(C) P-3, Q-5, R-4, S-1
(D) P-3, Q-2, R-5, S-4

CCN188

7. Considering the reaction sequence given below, the correct statement(s) is(are) **[JEE (Advanced) 2022]**

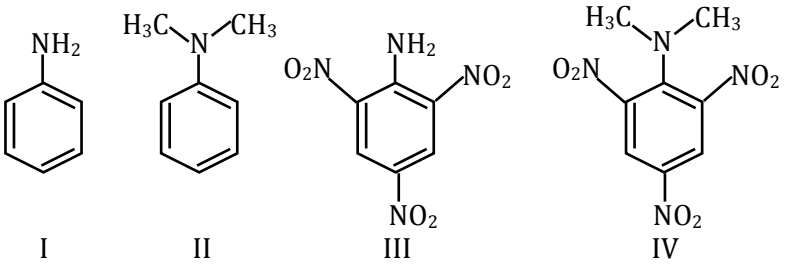


(A) **P** can be reduced to a primary alcohol using $NaBH_4$.
(B) Treating **P** with conc. NH_4OH solution followed by acidification gives **Q**.
(C) Treating **Q** with a solution of $NaNO_2$ in aq. HCl liberates N_2.
(D) **P** is more acidic than CH_3CH_2COOH.

CCN094

For More Join: @IITJEE_Advanced

8. Consider the following four compounds **I**, **II**, **III**, and **IV**.



Choose the correct statement(s). **[JEE (Advanced) 2020]**

(A) The order of basicity is **II** > **I** > **III** > **IV**.

(B) The magnitude of pK_b difference between **I** and **II** is more than that between **III** and **IV**.

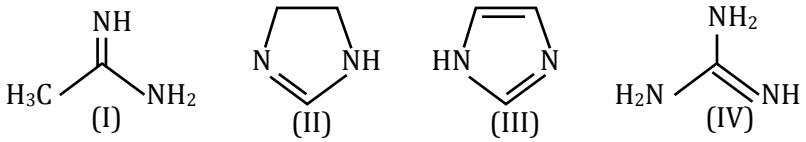
(C) Resonance effect is more in **III** than in **IV**.

(D) Steric effect makes compound **IV** more basic than **III**.

CCN091

9. The order of basicity among the following compounds is

[JEE (Advanced) 2017]



(A) II > I > IV > III

(B) IV > II > III > I

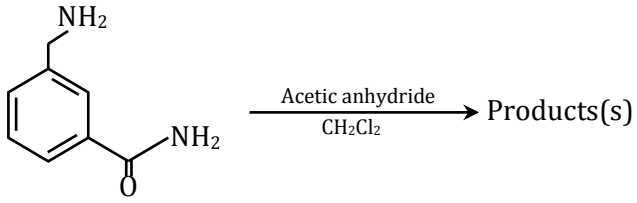
(C) I > IV > III > II

(D) IV > I > II > III

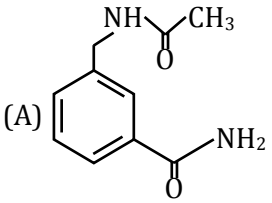
Rankers Academy JEE

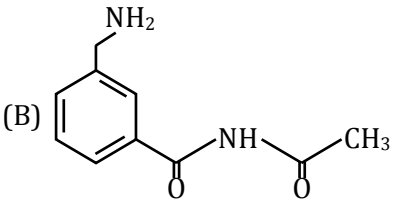
CCN090

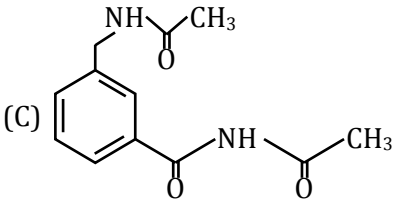
10. In the reaction shown below, the major product(s) formed is / are :

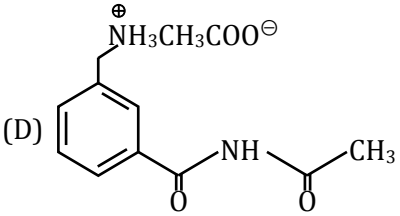


[JEE (Advanced) 2014]

(A)  + CH_3COOH

(B)  + CH_3COOH

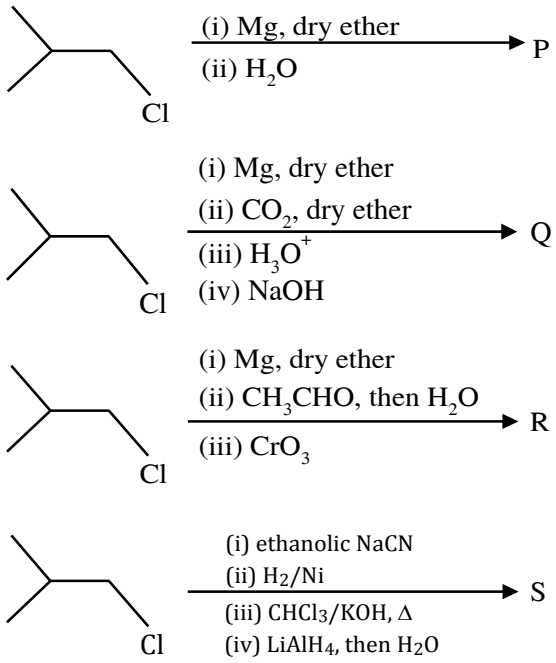
(C)  + H_2O

(D) 

CCN086

For More Join: @IITJEE_Advanced

11. In the following reactions, **P**, **Q**, **R**, and **S** are the major products.



The correct statement about **P**, **Q**, **R**, and **S** is **[JEE (Advanced) 2023]**

(A) **P** is a primary alcohol with four carbons.

(B) **Q** undergoes Kolbe's electrolysis to give an eight-carbon product.

(C) **R** has six carbons and it undergoes Cannizzaro reaction.

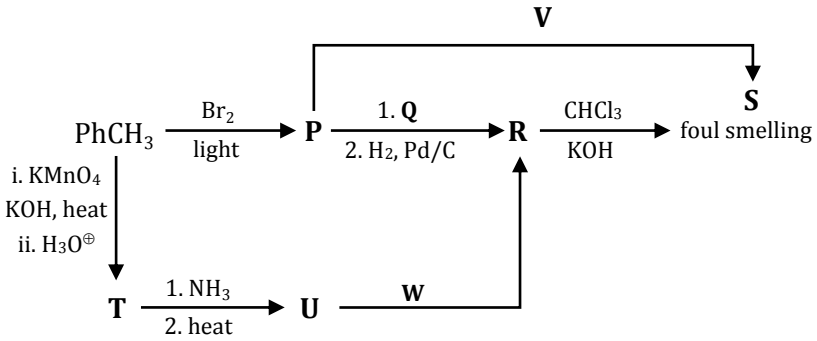
(D) **S** is a primary amine with six carbons.

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CCN147

12. Correct option(s) for the following sequence of reactions is(are)

[JEE (Advanced) 2021]



(A) **Q** = KNO_2, **W** = $LiAlH_4$

(B) **R** = benzenamine, **V** = KCN

(C) **Q** = $AgNO_2$, **R** = phenylmethanamine

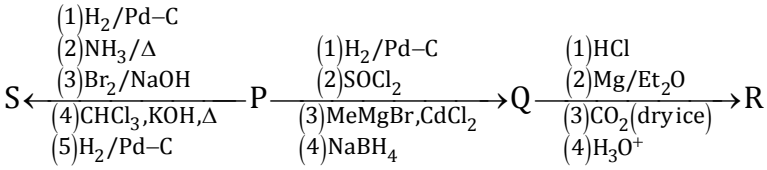
(D) **W** = $LiAlH_4$, **V** = AgCN

CCN093

For More Join: @IITJEE_Advanced

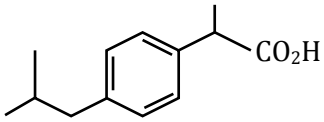
Paragraph for Question No. 13 to 14

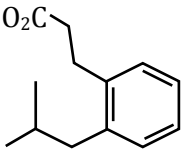
An organic acid P ($C_{11}H_{12}O_2$) can easily be oxidized to a dibasic acid which reacts with ethyleneglycol to produce a polymer dacron. Upon ozonolysis, P gives an aliphatic ketone as one of the products. P undergoes the following reaction sequences to furnish R via Q. The compound P also undergoes another set of reactions to produce S.

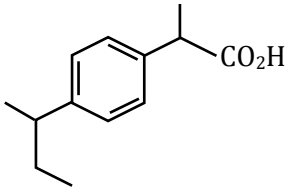


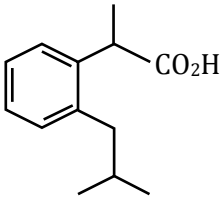
(There are two questions based on PARAGRAPH "A", the question given below is one of them)

13. The compound R is **[JEE (Advanced) 2018]**

(A) 

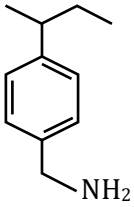
(B) 

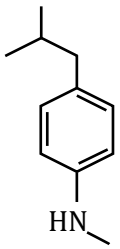
(C) 

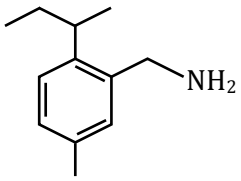
(D) 

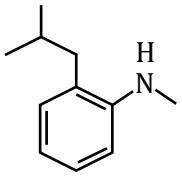
CCN145

14. The compound S is **[JEE (Advanced) 2018]**

(A) 

(B) 

(C) 

(D) 

CCN146

Paragraph for Question No. 15 to 16

A trinitro compound, 1, 3,5 tris-(4-nitrophenyl) benzene, on complete reaction with an excess of Sn/HCl gives major product, which on treatment with an excess of $NaNO_2$/HCl at 0°C provides **P** as the product. **P**, upon treatment with excess of H_2O at room temperature, gives the product **Q.** Bromination of **Q** in aqueous medium furnishes the product **R.** The compound **P** upon treatment with an excess of phenol under basic conditions gives the product **S.**

The molar mass difference between compounds **Q** and **R** is 474 mol^{-1} and between compounds **P** and **S** is 172.5 g mol^{-1}.

15. The number of heteroatoms present in one molecule of **R** is ____. **[JEE (Advanced) 2023]**

[Use: Molar mass (in g mol^{-1}): H = 1, C = 12, N = 14, O = 16, Br = 80, Cl = 35.5

Atoms other than C and H are considered as heteroatoms]

CCN148

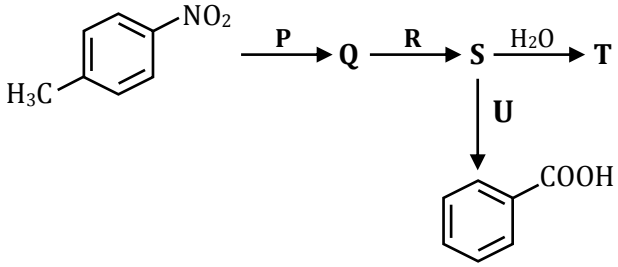
Rankers Academy JEE

For More Join: @IITJEE_Advanced

16. The total number of carbon atoms and heteroatoms present in one molecule of **S** is ____.
[Use: Molar mass in g mol^{-1}]: H = 1, C = 12, N = 14, O = 16, Br = 80, Cl = 35.5
Atoms other than C and H are considered as heteroatoms **[JEE (Advanced) 2023]**

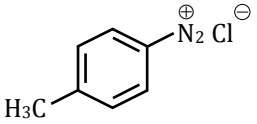
CCN149

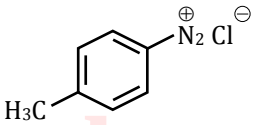
17. Consider the following reaction sequence, **[JEE (Advanced) 2022]**

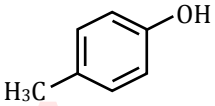


the correct option(s) is(are)

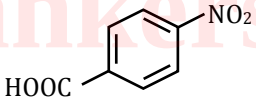
(A) **P** = H_2/Pd, ethanol **R** = $NaNO_2$/HCl **U** = 1. H_3PO_2
2. $KMnO_4$ - KOH, heat

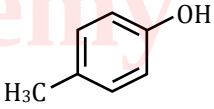
(B) **P** = Sn/HCl **R** = HNO_2 **S** = 

(C) **S** = 

T = 

U = 1. CH_3CH_2OH
2. $KMnO_4$ - KOH, heat

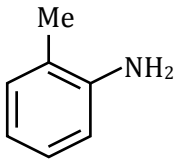
(D) **Q** = 

R = H_2/Pd, ethanol **T** = 

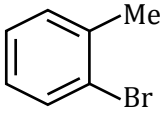
Rankers Academy JEE

CCN095

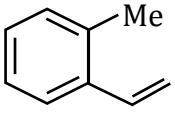
18. The reaction sequence(s) that would lead to *o*-xylene as the major product is (are)
[JEE (Advanced) 2021]

(A) 

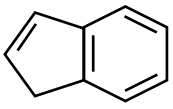
1. $NaNO_2$/HCl
273 K
2. CuCN
3. DIBAL-H
then $H_3O^{\oplus}$
4. N_2H_4, KOH
heat

(B) 

1. Mg, CO_2, $H_3O^{\oplus}$
2. $SOCl_2$
3. H_2, Pd-$BaSO_4$
4. Zn-Hg, HCl

(C) 

1. i. BH_3
ii. H_2O_2, NaOH
2. PBr_3
3. Zn, dil. HCl

(D) 
1. O_3, Zn/H_2O
2. N_2H_4, KOH, heat

CCN092

19. When benzamide is treated with $POCl_3$, the product is: **[IIT- 2004]**

(A) Benzonitrile (B) Aniline (C) Chlorobenzene (D) Benzylamine

CCN084

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	D	B	C	A	B	D	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	B	B	C	A	B	D	B	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	CD	CD	ABC	AC	AB	AC	ABC	B	B	B
Que.	11	12	13	14						
Ans.	C	D	B	A						

EXERCISE (S)

1.	4	**2.**	4	**3.**	4	**4.**	3	**5.**	2
6.	5	**7.**	6	**8.**	3	**9.**	9	**10.**	2
11.	3	**12.**	5	**13.**	6	**14.**	1.50	**15.**	2.50
16.	0.50	**17.**	12.50	**18.**	4.5				

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	93	1	80	4	3	8	3	45	9
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	2	3	3	4	3	4	2	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	5	1	5	1	4	3	3	4	2	95
Que.	31	32	33	34	35	36				
Ans.	3	2	20	3	2	1				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	C	A	C	B,C,D	CD	D	A
Que.	11	12	13	14	15	16	17	18	19	
Ans.	B	CD	A	B	9.00	51.00	ABC	AB	A	

For More Join: @IITJEE_Advanced